
Security & Middleware Lab (Practical)

Welcome to the practical lab for Security & Middleware! In this session, you will secure your database credentials using a `.env` file, configure **CORS middleware** in FastAPI to seamlessly accept requests from your React frontend, and verify your complete backend architecture via the interactive Swagger documentation.

1. Concept Overview

In this lab, you will:

- Install `python-dotenv` to load sensitive database credentials securely from an environment file.
 - Create a `.env` configuration file to keep passwords out of your source code.
 - Add `CORSMiddleware` to your FastAPI application to allow communication with your Vite React frontend (`http://localhost:5173`).
 - Test your secure backend configuration using the interactive Swagger UI (`/docs`).
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2. Step-by-Step Lab Instructions

Step 1: Install `python-dotenv`

Open your terminal in your project root, make sure your virtual environment is active, and install `python-dotenv` :

```
pip install python-dotenv
```

Step 2: Create a `.env` Configuration File

Create a new file named `.env` in the root directory of your project (same level as `main.py`) and store your database credentials:

```
env
# .env file
DB_USER=root
DB_PASSWORD=your_actual_mysql_password
```

```
DB_HOST=localhost
DB_PORT=3306
DB_NAME=ml_backend_db
```

Reminder: Never commit your `.env` file to public version control repositories like GitHub! (Add it to your `.gitignore` file).

Step 3: Update `database.py` to Use Environment Variables

Modify `database.py` to load environment variables and build the database URL dynamically:

```
# database.py
import os
from sqlalchemy import create_engine
from sqlalchemy.orm import DeclarativeBase, sessionmaker
from dotenv import load_dotenv

# Load environment variables from .env file
load_dotenv()

DB_USER = os.getenv("DB_USER")
DB_PASSWORD = os.getenv("DB_PASSWORD")
DB_HOST = os.getenv("DB_HOST", "localhost")
DB_PORT = os.getenv("DB_PORT", "3306")
DB_NAME = os.getenv("DB_NAME", "ml_backend_db")

# Construct SQLAlchemy database URL securely
DATABASE_URL = f"mysql+pymysql://{DB_USER}:{DB_PASSWORD}@{DB_HOST}:{DB_PORT}/{DB_NAME}"

engine = create_engine(DATABASE_URL)
SessionLocal = sessionmaker(autocommit=False, autoflush=False,
                             bind=engine)

class Base(DeclarativeBase):
    pass

def get_db():
    db = SessionLocal()
    try:
        yield db
    finally:
        db.close()
```

Step 4: Configure CORS Middleware in `main.py`

Update your root `main.py` file to import and add `CORSMiddleware`, allowing your React frontend (`http://localhost:5173`) to communicate with your backend without browser security blocks:

```
# main.py
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from database import engine, Base
from routers import predictions

# Automatically create database tables
Base.metadata.create_all(bind=engine)

app = FastAPI(title="Secure ML Full-Stack Backend", version="1.0")

# Define allowed frontend origins
origins = [
    "http://localhost:5173",
    "http://127.0.0.1:5173",
]

# Add CORS middleware to the FastAPI application
app.add_middleware(
    CORSMiddleware,
    allow_origins=origins,
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

# Include modular routers
app.include_router(predictions.router)

@app.get("/")
def read_root():
    return {"message": "Secure FastAPI backend with CORS and MySQL is running!"}
```

Step 5: Run and Test Your Secure Backend

1. Start your FastAPI server:

```
uvicorn main:app --reload
```

2. Open your browser and navigate to `[http://127.0.0.1:8000/docs]` (`http://127.0.0.1:8000/docs`) .
 3. Test your CRUD endpoints through Swagger UI. Your application is now securely connected to MySQL using environment variables and fully equipped with CORS headers for your React frontend!
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3. Reference Links & Further Reading

- **FastAPI CORS Documentation:** Official guide on configuring allowed origins and headers.
 - **Python-Dotenv Repository:** Best practices for managing secret configuration keys in Python.
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